Algebra II Rational Functions Cheatsheet

Module 5

Definition

$$f(x) = \frac{p(x)}{q(x)}, \qquad q(x) \neq 0$$

Domain

Exclude values that make the original denominator zero.

$$q(x) = 0 \Rightarrow x \text{ is not in the domain}$$

Holes and Vertical Asymptotes

Factor first.

$$\frac{(x-2)(x+5)}{(x-2)(x-3)}$$

Canceled factor:

$$x = 2 \Rightarrow \text{hole}$$

Remaining denominator zero:

$$x = 3 \Rightarrow \text{vertical asymptote}$$

Intercepts

$$x\text{-intercepts: zeros of the simplified numerator}$$

$$y\text{-intercept: } f(0) \text{ if defined}$$

Horizontal and Slant Asymptotes

For

$$f(x) = \frac{p(x)}{q(x)}$$

$$\deg p < \deg q \Rightarrow y = 0$$

$$\deg p = \deg q \Rightarrow y = \frac{\text{leading coefficient of } p}{\text{leading coefficient of } q}$$

$$\deg p > \deg q \Rightarrow \text{use polynomial division}$$

$$\deg p = \deg q + 1 \Rightarrow \text{slant asymptote}$$

Rational Equations

Before clearing denominators, state restrictions.

$$\frac{A}{B} = \frac{C}{D}, \qquad B \neq 0,\ D \neq 0$$

Then multiply by the LCD and solve.

Sign Chart

Critical values come from zeros and vertical asymptotes. Test one value in each interval to determine positive and negative behavior.

Quick Checks

- Domain restrictions come from the original denominator.
- Holes come from canceled factors.
- Vertical asymptotes come from uncanceled denominator factors.
- Check solutions to rational equations against restrictions.